

Bias reduction and uniform inference for additive regression

Kim Donghyun
Dept. of Statistics, SNU

June 19, 2026

Introduction

- Conventional nonparametric estimators have asymptotic bias
e.g. Local linear estimator with $d = 1$, $h \asymp n^{-1/5}$,
 $\|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\| = O_p(n^{-2/5})$, $\int_0^1 \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\|^2 = O_p(n^{-4/5})$, and
 $\sup_{x \in [0,1]} \|\hat{\mathbf{m}}(x) - \mathbf{m}(x)\| = O_p(n^{-2/5} \sqrt{\log n})$
- Existing bias reduction methods
 - undersmoothing
 - oversmoothing
Rely on bandwidth selection
 - Parametric help [Lee et al., 2020, Hong et al., 2023]
Rely on choice of assistance function
 - Least Square method [Cheng et al., 2018]

Local linear SBF estimator

Let Y be a response variable and $\mathbf{X} = (X_1, \dots, X_d)^\top$ be a covariate vector taking values in $[0, 1]^d$.

We consider the following additive model:

$$Y = m_0 + \sum_{j=1}^d m_j(X_j) + \sigma(\mathbf{X}) \cdot \varepsilon$$

with $\mathbb{E}[m_j(X_j)] = 0$.

$$\mathbf{m}(\mathbf{u}) \simeq \mathbf{m}(\mathbf{x}) + \sum_{j=1}^d (u_j - x_j) \frac{\partial m_j}{\partial x_j}(x_j) \quad \text{for } \mathbf{u} \simeq \mathbf{x}.$$

Local linear SBF estimator

The estimator $\hat{\mathbf{m}}^{\text{F,tp}} := (\bar{Y} + \hat{m}_1 + \cdots + \hat{m}_d, \hat{m}_{1,1}, \dots, \hat{m}_{1,d})^{\text{tp}} : [0, 1]^d \rightarrow \mathbb{R}^{d+1}$ is defined as the minimizer of

$$\mathcal{L}(\mathbf{f}^{\text{F,tp}}) = \int_{[0,1]^d} \left[n^{-1} \sum_{i=1}^n \left\{ Y_i - f_+(\mathbf{x}) - \sum_{j=1}^d \left(\frac{X_{ij} - x_j}{h_j} \right) f_{1,j}(\mathbf{x}) \right\}^2 \cdot \prod_{j=1}^d K_{h_j}(x_j, X_{ij}) \right] d\mathbf{x}$$

over the space of tuples $\mathbf{f}^{\text{F,tp}} = (f_+, f_{1,1}, \dots, f_{1,d})^\top$ where $f_+(\mathbf{x}) = f_1(x_1) + \cdots + f_d(x_d)$ for some univariate functions $f_j : [0, 1] \rightarrow \mathbb{R}$.

Local linear SBF estimator

Then, if $\hat{\mathbf{M}}_{jj}(x_j)$ is invertible for all $1 \leq j \leq d$ and $x_j \in [0, 1]$, the estimator $\hat{\mathbf{m}}^{\text{F,tp}}$ satisfies the following system of integral equations:

$$\hat{\mathbf{m}}_j^{\text{tp}}(x_j) = \tilde{\mathbf{m}}_j^{\text{tp}}(x_j) - \bar{Y} \cdot \mathbf{u}_0 - \sum_{k \neq j} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \hat{\mathbf{m}}_k^{\text{tp}}(x_k) dx_k, \quad 1 \leq j \leq d$$

where $\hat{\mathbf{m}}_j^{\text{tp}} := (\hat{m}_j, \hat{m}_{1,j})^\top : [0, 1] \rightarrow \mathbb{R}^2$ for $1 \leq j \leq d$,

$$\hat{\mathbf{M}}_{jj}(x_j) := n^{-1} \sum_{i=1}^n \text{vec} \left(\frac{X_{ij} - x_j}{h_j} \right) \text{vec} \left(\frac{X_{ij} - x_j}{h_j} \right)^\top K_{h_j}(x_j, X_{ij}),$$

$$\hat{\mathbf{M}}_{jk}(x_j, x_k) := n^{-1} \sum_{i=1}^n \text{vec} \left(\frac{X_{ij} - x_j}{h_j} \right) \text{vec} \left(\frac{X_{ik} - x_k}{h_k} \right)^\top K_{h_j}(x_j, X_{ij}) K_{h_k}(x_k, X_{ik}).$$

And $\tilde{\mathbf{m}}_j^{\text{tp}}(x_j)$ denotes the marginal local linear estimator of $\mathbb{E}(Y \mid X_j = x_j)$ as

$$\tilde{\mathbf{m}}_j^{\text{tp}}(x_j) := \hat{\mathbf{M}}_{jj}(x_j)^{-1} \cdot n^{-1} \sum_{i=1}^n \text{vec} \left(\frac{X_{ij} - x_j}{h_j} \right) K_{h_j}(x_j, X_{ij}) Y_i.$$

Local linear SBF estimator

According to [Jeon et al., 2022], under mild conditions, with probability tending to one, there exists a unique tuple $\hat{\mathbf{m}}^{\text{F, tp}} = (\bar{Y} + \hat{m}_1 + \cdots + \hat{m}_d, \hat{m}_{1,1}, \dots, \hat{m}_{1,d})^\top$ satisfying the SBF equations.

We impose the following constraints

$$\mathbf{u}_0^\top \cdot \int_0^1 \hat{\mathbf{M}}_{jj}(x_j) \hat{\mathbf{m}}_j^{\text{tp}}(x_j) dx_j = 0 \quad \text{for } 1 \leq j \leq d.$$

to identify each component function, which is consistent with following algorithm

$$\begin{aligned} \hat{\mathbf{m}}_j^{[\ell], \text{tp}}(x_j) &= \tilde{\mathbf{m}}_j^{\text{tp}}(x_j) - \bar{Y} \cdot \mathbf{u}_0 \\ &\quad - \sum_{k \leq j-1} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \hat{\mathbf{m}}_k^{[\ell], \text{tp}}(x_k) dx_k \\ &\quad - \sum_{k \geq j+1} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \hat{\mathbf{m}}_k^{[\ell-1], \text{tp}}(x_k) dx_k. \end{aligned}$$

Local linear SBF estimator

To analyze the convergence of the SBF algorithm and to formulate a stopping rule, we define a semi-norm for the full tuples $\mathbf{f}^{\text{F,tp}} = (f_+, f_{1,1}, \dots, f_{1,d})^\top$ $f_j : [0, 1] \rightarrow \mathbb{R}$.

For $\mathbf{x} \in [0, 1]^d$, let $\hat{p}(\mathbf{x}) := n^{-1} \sum_{i=1}^n \prod_{j=1}^d K_{h_j}(x_j, X_{ij})$ and

$\hat{p}_j(x_j) := n^{-1} \sum_{i=1}^n K_{h_j}(x_j, X_{ij})$.

Using these kernel density estimators, define the semi-norm $\|\cdot\|_{2,n}$ by

$$\|\mathbf{f}^{\text{F,tp}}\|_{2,n}^2 := \int_{[0,1]^d} |f_+(\mathbf{x})|^2 \hat{p}(\mathbf{x}) d\mathbf{x} + \sum_{j=1}^d \int_0^1 |f_{1,j}(x_j)|^2 \hat{p}_j(x_j) dx_j.$$

By Theorem 1 of [Jeon et al., 2022], under mild regularity conditions, the following holds with probability tending to one: if the initial estimator satisfies $\|\hat{\mathbf{m}}^{[0],\text{F,tp}}\|_{2,n} \leq C'$ for some constant $0 < C' < \infty$, then there exist constants $0 < C < \infty$ and $0 < \gamma < 1$ such that $\|\hat{\mathbf{m}}^{[\ell],\text{F,tp}} - \hat{\mathbf{m}}^{\text{F,tp}}\|_{2,n} \leq C\gamma^\ell$ for all $\ell \geq 1$.

Bias reduction method

Bias reduction for nonparametric and semiparametric regression models, Cheng et al., 2018

- Let $\hat{m}_h(x)$ denote the local linear estimator using the bandwidth h .
- For local linear estimator, $\mathbb{E}(\hat{m}_h(x) \mid \mathbb{X}) - m(x) = \frac{1}{2}\mu_2 m''(x)h^2 + o_p(h^2)$.
- To reduce the deterministic term, consider a linear regression model for

$$\hat{m}_h(x) = \alpha(x) + \beta(x)h^2 + \tilde{\sigma}(h)(x)$$

where $\tilde{\sigma}(h) = \text{Var}(\hat{m}_h(x))$.

- Compute least squares estimators using $(\hat{m}_{h_1}(x), h_1^2), \dots, (\hat{m}_{h_L}(x), h_L^2)$
- Consider $\tilde{m}^{\text{BR}}(x) = \hat{\alpha}(x)$, $\hat{m}^{\text{BR}}(x) = \hat{m}_h(x) - \hat{\beta}(x)h^2$ as bias-reduced estimators.

Asymptotic expansion of local linear SBF estimator

Assume following conditions

- The joint density p is bounded away from zero and infinity on $[0, 1]^d$. For each $1 \leq j \leq d$, the derivative p'_j exists and is continuous on $[0, 1]$. Also, for each $1 \leq j \neq k \leq d$, the partial derivatives $\frac{\partial}{\partial x_j} p_{jk}(x_j, x_k)$, $\frac{\partial}{\partial x_k} p_{jk}(x_j, x_k)$ exist and are continuous on $[0, 1]^2$.
- The baseline kernel K is symmetric, bounded, Lipschitz continuous, supported on $[-1, 1]$ and satisfies $\int K(u) du = 1$.
- The random error ε satisfies $\mathbb{E}[|\varepsilon|^\alpha] < \infty$ for some $\alpha > 10/3$. Also, its conditional second moment function $E(\varepsilon^2 \mid \mathbf{X} = \cdot)$ is bounded on $[0, 1]^d$.
- For $1 \leq j \leq d$, the true additive component m_j is three times continuously differentiable on $[0, 1]$.
- There exist constants $c_j > 0$ with $c_j + c_k < (\alpha - 2)/\alpha$ for all $1 \leq j \neq k \leq d$, where α is the constant in (A4), such that $h_j = o(1)$, $n^{c_j} h_j \gtrsim 1$ and $h_k^2 \lesssim h_j$ for all $1 \leq j \neq k \leq d$.

Asymptotic expansion of local linear SBF estimator

Proposition 1

Assume (A1)–(A5). Then for each $1 \leq j \leq d$,

$$\begin{aligned} \sup_{x_j \in [0,1]} \left| \hat{\mathbf{m}}_j^{\text{tp}}(x_j) - \mathbf{m}_j^{\text{tp}}(x_j) - \tilde{\mathbf{m}}_j^{A,\text{tp}}(x_j) - \frac{1}{2} h_j^2 \mathbf{N}_j(x_j)^{-1} \boldsymbol{\gamma}_j(x_j) m_j''(x_j) \right| \\ = O_p \left(\sum_{k=1}^d h_k^3 + \sqrt{\frac{\log n}{n}} \right). \end{aligned}$$

Asymptotic expansion of local linear SBF estimator

where

$$\mathbf{N}_j(x_j) := \int_0^1 \mathbf{vec}\left(\frac{v - x_j}{h_j}\right) \mathbf{vec}\left(\frac{v - x_j}{h_j}\right)^\top K_{h_j}(x_j, v) \, dv,$$

$$\gamma_j(x_j) := \int_0^1 \left(\frac{v - x_j}{h_j}\right)^2 \mathbf{vec}\left(\frac{v - x_j}{h_j}\right) K_{h_j}(x_j, v) \, dv.$$

$$\tilde{\mathbf{m}}_j^{A, \text{tp}}(x_j) = n^{-1} \sum_{i=1}^n \hat{\mathbf{M}}_{jj}(x_j)^{-1} \mathbf{vec}\left(\frac{X_{ij} - x_j}{h_j}\right) K_{h_j}(x_j, X_{ij}) \cdot \varepsilon_i, \quad x_j \in [0, 1].$$

For $x_j \in [2h_j, 1 - 2h_j]$, $\mathbf{N}_j(x_j) \equiv \mathbf{N}$ and $\gamma_j(x_j) \equiv \gamma$.

Asymptotic expansion of local linear SBF estimator

Specifically, if we further assume that $h_j \asymp h$ for some $h > 0$, we obtain the following convergence rates:

- (i) Pointwise convergence: for each $x_j \in (0, 1)$, $|\hat{m}_j(x_j) - m_j(x_j)| = O_p(h_j^2 + (nh_j)^{-1/2})$;
- (ii) L_2 convergence: $\int_0^1 |\hat{m}_j(x_j) - m_j(x_j)|^2 dx_j = O_p(h_j^4 + (nh_j)^{-1})$;
- (iii) Uniform convergence: $\sup_{x_j \in [0, 1]} |\hat{m}_j(x_j) - m_j(x_j)| = O_p(h_j^2 + \sqrt{\frac{\log n}{nh_j}})$.

Sketch of proof of Proposition 1

For $1 \leq j \neq k \leq d$, let

$$\begin{aligned}\tilde{\mathbf{m}}_j^{B,\text{tp}}(x_j) &:= \sum_{i=1}^n \hat{\kappa}_{ij}(x_j) \cdot \left(m_j(X_{ij}) - \text{vec} \left(\frac{X_{ij} - x_j}{h_j} \right)^\top \mathbf{m}_j^{\text{tp}}(x_j) \right) \\ \tilde{\mathbf{m}}_{jk}^{C,\text{tp}}(x_j) &:= \sum_{i=1}^n \int_0^1 \hat{\mathbf{M}}_{jj}(x_j) \hat{\kappa}_{ij}(x_j) K_{h_k}(x_k, X_{ik}) \\ &\quad \cdot \left(m_k(X_{ik}) - \text{vec} \left(\frac{X_{ik} - x_k}{h_k} \right)^\top \mathbf{m}_k^{\text{tp}}(x_k) \right) dx_k\end{aligned}$$

Then, the SBF equations can be written as

$$\begin{aligned}\hat{\mathbf{m}}_j^{\text{tp}}(x_j) &= \mathbf{m}_j^{\text{tp}}(x_j) + (m_0 - \bar{Y}) \cdot \mathbf{u}_0 + \tilde{\mathbf{m}}_j^{A,\text{tp}}(x_j) + \tilde{\mathbf{m}}_j^{B,\text{tp}}(x_j) + \sum_{k \neq j} \hat{\mathbf{M}}_{jj}(x_j)^{-1} \tilde{\mathbf{m}}_{jk}^{C,\text{tp}}(x_j) \\ &\quad - \sum_{k \neq j} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \left(\hat{\mathbf{m}}_k^{\text{tp}}(x_k) - \mathbf{m}_k^{\text{tp}}(x_k) \right) dx_k.\end{aligned}$$

Sketch of proof of Proposition 1

This also implies

$$\begin{aligned} \hat{\Delta}_j^{\text{tp}}(x_j) + \sum_{k \neq j} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \hat{\Delta}_k^{\text{tp}}(x_k) dx_k \\ = (m_0 - \bar{Y}) \cdot \mathbf{u}_0 + \hat{I}_{1,j}(x_j) + \sum_{k \neq j} \left(\hat{I}_{2,jk}(x_j) + \hat{I}_{3,jk}(x_j) \right), \end{aligned}$$

where

$$\begin{aligned} \hat{I}_{1,j}(x_j) &:= \tilde{\mathbf{m}}_j^{B,\text{tp}}(x_j) - \frac{h_j^2}{2} \mathbf{N}_j(x_j)^{-1} \boldsymbol{\gamma}_j(x_j) m_j''(x_j), \\ \hat{I}_{2,jk}(x_j) &:= \hat{\mathbf{M}}_{jj}(x_j)^{-1} \tilde{\mathbf{m}}_{jk}^{C,\text{tp}}(x_j) \\ &\quad - \frac{h_k^2}{2} \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \mathbf{N}_k(x_k)^{-1} \boldsymbol{\gamma}_k(x_k) m_k''(x_k) dx_k, \\ \hat{I}_{3,jk}(x_j) &:= \int_0^1 \hat{\mathbf{M}}_{jj}(x_j)^{-1} \hat{\mathbf{M}}_{jk}(x_j, x_k) \tilde{\mathbf{m}}_k^{A,\text{tp}}(x_k) dx_k. \end{aligned}$$

Sketch of proof of Proposition 1

$$\sup_{x_j \in [0,1]} |\hat{I}_{1,j}(x_j)| = O_p\left(h_j^3 + \sqrt{\frac{\log n}{nh_j}} h_j^2\right),$$

$$\sup_{x_j \in [0,1]} |\hat{I}_{2,jk}(x_j)| = O_p\left(h_k^3 + h_k^2 h_j + \sqrt{\frac{\log n \cdot h_k^2}{nh_j}}\right),$$

$$\sup_{x_j \in [0,1]} |\hat{I}_{3,jk}(x_j)| = O_p\left(\sqrt{\frac{\log n}{n}}\right).$$

Bias reduction

To specify the dependence of the bandwidths, let $\hat{m}_j^{\text{tp}}(\cdot; \mathbf{h})$ denote the local linear SBF estimator based on the bandwidth vector $\mathbf{h} = (h_1, \dots, h_d)^\top$. We then define the corresponding estimator of m_j as $\hat{m}_j(x_j; \mathbf{h}) := \mathbf{u}_0^\top \cdot \hat{m}_j^{\text{tp}}(x_j; \mathbf{h})$ for all $x_j \in [0, 1]$. Proposition 1 tells that

$$\hat{m}_j(x_j; \mathbf{h}) = m_j(x_j) + \beta_j(x_j) \cdot h_j^2 + n^{-1/2} h_j^{-1/2} \cdot \zeta_j(x_j) + O_p\left(\sum_{k=1}^d h_k^3 + \sqrt{\frac{\log n}{n}}\right),$$

where $\beta_j(x_j) := \frac{1}{2} \mathbf{u}_0^\top \mathbf{N}_j(x_j)^{-1} \boldsymbol{\gamma}_j(x_j) m_j''(x_j)$ and $n^{-1/2} h_j^{-1/2} \cdot \zeta_j(x_j) = O_p(1)$, pointwise. For interior x_j , $\beta_j(x_j) = \frac{1}{2} \mathbf{u}_0^\top \mathbf{N}^{-1} \boldsymbol{\gamma} \cdot m_j''(x_j)$.

Motivated by this expression, given bandwidth vectors for $\ell = 1, \dots, L$, we fit the following linear regression model:

$$\hat{m}_j(x_j; \mathbf{h}_\ell) = \alpha_j(x_j) + \beta_j(x_j) h_{\ell j}^2 + n^{-1/2} h_j^{-1/2} \cdot \zeta_j(x_j), \quad \ell = 1, \dots, L,$$

where $\alpha_j(x_j) := m_j(x_j)$.

Bias reduction

The least squares estimates of $\alpha_j(x_j)$ and $\beta_j(x_j)$ are given by

$$\hat{\alpha}_{j,L}(x_j) := \sum_{\ell=1}^L \left\{ \frac{\sum_{\ell'=1}^L h_{\ell'j}^4 - (\sum_{\ell'=1}^L h_{\ell'j}^2) h_{\ell j}^2}{L \cdot \sum_{\ell'=1}^L h_{\ell'j}^4 - (\sum_{\ell'=1}^L h_{\ell'j}^2)^2} \right\} \cdot \hat{m}_j(x_j; \mathbf{h}_\ell),$$
$$\hat{\beta}_{j,L}(x_j) := \sum_{\ell=1}^L \left\{ \frac{L \cdot h_{\ell j}^2 - (\sum_{\ell'=1}^L h_{\ell'j}^2)}{L \cdot \sum_{\ell'=1}^L h_{\ell'j}^4 - (\sum_{\ell'=1}^L h_{\ell'j}^2)^2} \right\} \cdot \hat{m}_j(x_j; \mathbf{h}_\ell).$$

Consequently, for $x_j \in [0, 1]$, we define the bias-reduced SBF estimators of $m_j(x_j)$ as

$$\tilde{m}_j^{\text{BR}}(x_j) := \hat{\alpha}_{j,L}(x_j) \quad \text{and} \quad \hat{m}_j^{\text{BR}}(x_j) := \hat{m}_j(x_j; \mathbf{h}) - \hat{\beta}_{j,L}(x_j) h_j^2.$$

Bias reduction

Alternatively, one may consider the weighted least squares estimators of $\alpha_j(x_j)$ and $\beta_j(x_j)$. We can employ weights proportional to $h_{\ell j}$ as

$$\hat{\alpha}_{j,L}^W(x_j) := \sum_{\ell=1}^L \left\{ \frac{h_{\ell j} \sum_{\ell'=1}^L h_{\ell' j}^5 - (\sum_{\ell'=1}^L h_{\ell' j}^3) h_{\ell j}^3}{(\sum_{\ell'=1}^L h_{\ell' j})(\sum_{\ell'=1}^L h_{\ell' j}^5) - (\sum_{\ell'=1}^L h_{\ell' j}^3)^2} \right\} \cdot \hat{m}_j(x_j; \mathbf{h}_{\ell}),$$
$$\hat{\beta}_{j,L}^W(x_j) := \sum_{\ell=1}^L \left\{ \frac{(\sum_{\ell'=1}^L h_{\ell' j}) \cdot h_{\ell j}^3 - (\sum_{\ell'=1}^L h_{\ell' j}^3) h_{\ell j}}{(\sum_{\ell'=1}^L h_{\ell' j})(\sum_{\ell'=1}^L h_{\ell' j}^5) - (\sum_{\ell'=1}^L h_{\ell' j}^3)^2} \right\} \cdot \hat{m}_j(x_j; \mathbf{h}_{\ell}).$$

Then, the weighted versions of the bias-reduced SBF estimators are given by

$$\tilde{m}_j^{\text{WBR}}(x_j) := \hat{\alpha}_{j,L}^W(x_j) \quad \text{and} \quad \hat{m}_j^{\text{WBR}}(x_j) = \hat{m}_j(x_j; \mathbf{h}) - \hat{\beta}_{j,L}^W(x_j) h_j^2.$$

Asymptotic properties of the bias-reduced estimators

- For $1 \leq \ell \leq L$ and $1 \leq j \neq k \leq d$, it holds that $h_{\ell j} \asymp h_{j,\text{BR}}$ and $h_{\ell k} \asymp h_{-j,\text{BR}}$. For $1 \leq j \leq d$, there exist constants $c'_j, c'_{-j} > 0$ with $c'_j + c'_{-j} < (\alpha - 2)/\alpha$, where α is the constant in (A4), such that $h_{j,\text{BR}} = o((\log n)^{-1})$, $n^{c'_j} h_{j,\text{BR}} \gtrsim 1$, $n^{c'_{-j}} h_{-j,\text{BR}} \gtrsim 1$ and $h_{-j,\text{BR}}^2 \lesssim h_{j,\text{BR}}$.

Asymptotic properties of the bias-reduced estimators

Theorem 1

Assume (A1)–(A6). Then, for $\# = \text{BR}$ and $\# = \text{WBR}$, the following convergence rates hold:

(i) Pointwise convergence:

$$|\tilde{m}_j^\#(x_j) - m_j(x_j)| = O_p(h_{j,\text{BR}}^3 + h_{-j,\text{BR}}^3 + n^{-1/2}h_{j,\text{BR}}^{-1/2}), \quad x_j \in I_j,$$

$$|\tilde{m}_j^\#(x_j) - m_j(x_j)| = O_p(h_{j,\text{BR}}^2 + h_{-j,\text{BR}}^3 + n^{-1/2}h_{j,\text{BR}}^{-1/2}), \quad x_j \in [0, 1] \setminus I_j.$$

(ii) L_2 convergence:

$$\int_{I_j} |\tilde{m}_j^\#(x_j) - m_j(x_j)|^2 dx_j = O_p(h_{j,\text{BR}}^6 + h_{-j,\text{BR}}^6 + n^{-1}h_{j,\text{BR}}^{-1}),$$

$$\int_0^1 |\tilde{m}_j^\#(x_j) - m_j(x_j)|^2 dx_j = O_p(h_{j,\text{BR}}^5 + h_{-j,\text{BR}}^6 + n^{-1}h_{j,\text{BR}}^{-1}).$$

Asymptotic properties of the bias-reduced estimators

Theorem 1

(iii) *Uniform convergence:*

$$\sup_{x_j \in I_j} |\tilde{m}_j^\#(x_j) - m_j(x_j)| = O_p \left(h_{j,\text{BR}}^3 + h_{-j,\text{BR}}^3 + \sqrt{\frac{\log n}{nh_{j,\text{BR}}}} \right),$$

$$\sup_{x_j \in [0,1]} |\tilde{m}_j^\#(x_j) - m_j(x_j)| = O_p \left(h_{j,\text{BR}}^2 + h_{-j,\text{BR}}^3 + \sqrt{\frac{\log n}{nh_{j,\text{BR}}}} \right).$$

Asymptotic properties of the bias-reduced estimators

- When the bandwidth vectors $\mathbf{h}$ and $\mathbf{h}_\ell$ satisfy $h_j = o(h_{j,\text{BR}})$, $h_{j,\text{BR}} = o(h_j^{2/3})$, and $h_{-j,\text{BR}} \asymp h_{-j}$, the pointwise convergence rate for $\tilde{m}_j^{\text{BR}}$ is strictly faster than that for $\hat{m}_j(\cdot; \mathbf{h})$ in the interior region.
- Specifically, with $h_j \asymp n^{-1/5}$ and $h_{\ell j} \asymp n^{-1/5}$, the leading bias of $\tilde{m}_j^{\text{BR}}$ is $O(n^{-3/5})$ in the interior region and $O(n^{-2/5})$ in the boundary region, pointwise.

Asymptotic properties of the bias-reduced estimators

Sketch of proof of Theorem 1

$$a_{\ell j} := \frac{\sum_{\ell'=1}^L h_{\ell'j}^4 - (\sum_{\ell'=1}^L h_{\ell'j}^2) h_{\ell j}^2}{L \cdot \sum_{\ell'=1}^L h_{\ell'j}^4 - (\sum_{\ell'=1}^L h_{\ell'j}^2)^2}.$$

By the definition, it holds that

$$\sum_{\ell=1}^L a_{\ell j} = 1, \quad \sum_{\ell=1}^L a_{\ell j} \cdot h_{\ell j}^2 = 0.$$

Asymptotic properties of the bias-reduced estimators

$$\begin{aligned}\tilde{m}_j^{\text{BR}}(x_j) &= m_j(x_j) + \sum_{\ell=1}^L a_{\ell j} \cdot \frac{1}{2} h_{\ell j}^2 \mathbf{u}_0^\top \mathbf{N}_j(x_j; h_{\ell j})^{-1} \boldsymbol{\gamma}_j(x_j; h_{\ell j}) m_j''(x_j) \\ &\quad + \sum_{\ell=1}^L a_{\ell j} \cdot \mathbf{u}_0^\top \tilde{\mathbf{m}}_j^{A, \text{tp}}(x_j; h_{\ell j}) + O_p\left(h_{j, \text{BR}}^3 + h_{-j, \text{BR}}^3 + \sqrt{\frac{\log n}{n}}\right),\end{aligned}$$

where the O_p term holds uniformly over $x_j \in [0, 1]$.

$$\begin{aligned}\sup_{x_j \in I_j} \left| \sum_{\ell=1}^L a_{\ell j} \cdot \frac{1}{2} h_{\ell j}^2 \mathbf{u}_0^\top \mathbf{N}_j(x_j; h_{\ell j})^{-1} \boldsymbol{\gamma}_j(x_j; h_{\ell j}) m_j''(x_j) \right| &= 0, \\ \sup_{x_j \in [0, 1]} \left| \sum_{\ell=1}^L a_{\ell j} \cdot \frac{1}{2} h_{\ell j}^2 \mathbf{u}_0^\top \mathbf{N}_j(x_j; h_{\ell j})^{-1} \boldsymbol{\gamma}_j(x_j; h_{\ell j}) m_j''(x_j) \right| &= O(h_{j, \text{BR}}^2).\end{aligned}$$

Asymptotic properties of the bias-reduced estimators

we can show that

$$\begin{aligned} |\mathbf{u}_0^\top \tilde{\mathbf{m}}_j^{A, \text{tp}}(x_j; h_{\ell j})| &= O_p(n^{-1/2} h_{j, \text{BR}}^{-1/2}), \quad x_j \in (0, 1), \\ \int_0^1 |\mathbf{u}_0^\top \tilde{\mathbf{m}}_j^{A, \text{tp}}(x_j; h_{\ell j})|^2 dx_j &= O_p(n^{-1} h_{j, \text{BR}}^{-1}), \\ \sup_{x_j \in [0, 1]} |\mathbf{u}_0^\top \tilde{\mathbf{m}}_j^{A, \text{tp}}(x_j; h_{\ell j})| &= O_p\left(\sqrt{\frac{\log n}{n h_{j, \text{BR}}}}\right). \end{aligned} \tag{1}$$

Asymptotic properties of the bias-reduced estimators

Theorem 2

Assume that (A1)–(A6) hold and that $h_j = o(h_{j,\text{BR}})$. Then, for $\# = \text{BR}$ and $\# = \text{WBR}$, the following convergence rates hold:

(i) Pointwise convergence:

$$|\hat{m}_j^\#(x_j) - m_j(x_j)| = O_p(h_j^2 h_{j,\text{BR}} + r_{nj} + n^{-1/2} h_j^{-1/2}), \quad x_j \in I_j,$$

$$|\hat{m}_j^\#(x_j) - m_j(x_j)| = O_p(h_j^2 + r_{nj} + n^{-1/2} h_j^{-1/2}), \quad x_j \in [0, 1] \setminus I_j.$$

(ii) L_2 convergence:

$$\int_{I_j} |\hat{m}_j^\#(x_j) - m_j(x_j)|^2 dx_j = O_p(h_j^4 h_{j,\text{BR}}^2 + r_{nj}^2 + n^{-1} h_j^{-1}),$$

$$\int_0^1 |\hat{m}_j^\#(x_j) - m_j(x_j)|^2 dx_j = O_p(h_j^4 h_{j,\text{BR}} + r_{nj}^2 + n^{-1} h_j^{-1}).$$

Asymptotic properties of the bias-reduced estimators

Theorem 2

(iii) *Uniform convergence:*

$$\sup_{x_j \in I_j} |\hat{m}_j^\#(x_j) - m_j(x_j)| = O_p \left(h_j^2 h_{j,\text{BR}} + r_{nj} + \sqrt{\frac{\log n}{nh_j}} \right),$$

$$\sup_{x_j \in [0,1]} |\hat{m}_j^\#(x_j) - m_j(x_j)| = O_p \left(h_j^2 + r_{nj} + \sqrt{\frac{\log n}{nh_j}} \right).$$

where

$$r_{nj} := h_j^2 \frac{h_{-j,\text{BR}}^3}{h_{j,\text{BR}}^2} + h_{-j}^3, \quad 1 \leq j \leq d.$$

Asymptotic properties of the bias-reduced estimators

- Assuming that $h_{-j,\text{BR}}^3 = o(h_{j,\text{BR}}^2)$, the leading biases of $\hat{m}_j^{\text{BR}}$ and $\hat{m}_j^{\text{WBR}}$ converge faster than that of $\hat{m}_j(\cdot; \mathbf{h})$ in the interior region.
- At the boundary, however, the biases of $\hat{m}_j^{\text{BR}}$, $\hat{m}_j^{\text{WBR}}$ and $\hat{m}_j(\cdot; \mathbf{h})$ have the same asymptotic magnitude.
- Suppose that $h_j \asymp n^{-1/5}$ and $h_{j,\text{BR}}, h_{-j,\text{BR}} \asymp n^{-1/5+\epsilon}$ for some constant $0 \leq \epsilon < 1/15$. Then pointwise convergence rates for $\tilde{m}_j^{\text{BR}}$ and $\hat{m}_j^{\text{BR}}$ at $x_j \in (0, 1)$ reduce to

$$|\tilde{m}_j^{\text{BR}}(x_j) - m_j(x_j)| = O_p(n^{-3/5+3\epsilon} + n^{-2/5-\epsilon/2}),$$

$$|\hat{m}_j^{\text{BR}}(x_j) - m_j(x_j)| = O_p(n^{-2/5}).$$

Comparison with parametrically-helped estimators

- Parametric help approach

For additive model, *Additive regression with parametric help*, Hong et al., 2023

- projects each additive component onto a working parametric model

Let $g_j : [0, 1] \rightarrow \mathbb{R}$ denote a prespecified, twice differentiable univariate function for $1 \leq j \leq d$. We define the parameter

$$\theta_{0j} := \operatorname{argmin}_{\theta_j \in \mathbb{R}} \mathbb{E} \left(m_j''(X_j) - \theta_j [g_j''(X_j) - \mathbb{E} g_j''(X_j)] \right)^2 = \frac{\operatorname{Cov}(m_j''(X_j), g_j''(X_j))}{\operatorname{Var}(g_j''(X_j))}.$$

we obtain the PH-SBF estimator $\hat{m}_j^{\text{PH}}$ that has an asymptotic bias of $h_j^2 \cdot \beta_j^{\text{PH}}$, where

$$\beta_j^{\text{PH}}(x_j) := \frac{1}{2} \mathbf{u}_0^\top \mathbf{N}_j(x_j)^{-1} \boldsymbol{\gamma}_j(x_j) \cdot \left(m_j''(x_j) - \theta_{0j} (g_j''(x_j) - \mathbb{E} g_j''(X_j)) \right).$$

by the definition of θ_{0j} , it holds that

$$\mathbb{E} \left(m_j''(X_j) - \theta_{0j} [g_j''(X_j) - \mathbb{E} g_j''(X_j)] \right)^2 \leq \mathbb{E} \left(m_j''(X_j) \right)^2.$$

Uniform distributional approximation using $\hat{m}_j^\#$

Assume following conditions further

- The density p satisfies (A1). Moreover, the density p is **twice?** continuously differentiable on $[0, 1]^d$.
- For some $\alpha' > 4$, it holds that $\mathbb{E}[|\varepsilon|^{\alpha'}] < \infty$ and $\mathbb{E}[|X_j|^{\alpha'}] < \infty$ for $j = 1, \dots, d$. Also, the conditional variance function $\sigma^2 := \mathbb{E}(\varepsilon^2 | \mathbf{X} = \cdot)$ is bounded away from zero and infinity on $[0, 1]^d$ and **has continuous second-order? partial derivatives on $[0, 1]^d$.**
- It holds that $h_j, h_{-j}, h_{-j, \text{BR}} \asymp n^{-b}$ for some $1/5 \leq b < (\alpha' - 2)/\alpha'$, where α' is the constant in (B2).

Uniform distributional approximation using $\hat{m}_j^\#$

For any closed interval $I \subset (0, 1)$, define

$$A_{nj}(I) := (2 \log(|I|/h_j))^{1/2}, \quad B_{nj}(I) := A_{nj}(I) + \frac{\log(\lambda_K/(4\pi^2))}{2A_{nj}(I)},$$

where $\lambda_K := \nu_0^{-1} \cdot \int \{K'(u)\}^2 du$. For $\# \in \{\text{BR}, \text{WBR}\}$, let

$$\hat{\Delta}_{nj}^\#(I) := \nu_0^{-1/2} \cdot \sup_{x_j \in I} \left| \{nh_j \sigma_j(x_j)^{-2} p_j(x_j)\}^{1/2} \{\hat{m}_j^\#(x_j) - m_j(x_j)\} \right|.$$

where $\nu_0 := \int_{-\infty}^{\infty} K^2(u) du$ and $\sigma_j^2 := \text{Var}(\varepsilon^2 | X_j = \cdot)$.

Theorem 3

Assume that (A2), (A4), (A6) and (B1)–(B3) hold and that $h_j = o(h_{j,\text{BR}})$. Then, for any closed interval $I \subset (0, 1)$ and $\# \in \{\text{BR}, \text{WBR}\}$,

$$\lim_{n \rightarrow \infty} \mathbb{P} \left(A_{nj}(I) \left\{ \hat{\Delta}_{nj}^\#(I) - B_{nj}(I) \right\} \leq u \right) = \exp(-2e^{-u}), \quad u \in \mathbb{R}.$$

Uniform Testing and Simultaneous Confidence Band

Let $m_{j0} : [0, 1] \rightarrow \mathbb{R}$ denote a prespecified function and $I \subset (0, 1)$ be a closed interval. We consider the following hypothesis:

$$H_0 : m_j(x_j) = m_{j0}(x_j) \text{ for all } x_j \in I \quad \text{vs} \quad H_1 : \text{not } H_0.$$

We define the test statistic $T_{nj}(I)$ by

$$T_{nj}(I) := A_{nj}(I) \left\{ \nu_0^{-1/2} \cdot \sup_{x_j \in I} \left| \{nh_j \hat{\sigma}_j(x_j)^{-2} \hat{p}_j(x_j)\}^{1/2} \{\hat{m}_j^\#(x_j) - m_{j0}(x_j)\} \right| - B_{nj}(I) \right\}.$$

The null hypothesis H_0 is rejected at a significance level $\alpha \in (0, 1)$ if

$$T_{nj}(I) > -\log \left(\frac{1}{2} \log \frac{1}{1 - \alpha} \right) =: C_\alpha.$$

Uniform Testing and Simultaneous Confidence Band

Accordingly, the p-value is given by

$$\text{p-value} := 1 - \exp \left(- 2 \exp(-T_{nj}(I)) \right).$$

Furthermore, we can construct a simultaneous confidence band (SCB) for m_j over $I \subset (0, 1)$ satisfying

$$\mathbb{P} \left(m_j(x_j) \in \text{SCB}(x_j; \alpha, I) \text{ for all } x_j \in I \right) \rightarrow 1 - \alpha.$$

Here, the SCB is defined by

$$\text{SCB}(x_j; \alpha, I) := [\hat{m}_j^{\text{BR}}(x_j) - \Delta_{1-\alpha}(x_j; I), \hat{m}_j^{\text{BR}}(x_j) + \Delta_{1-\alpha}(x_j; I)], \quad x_j \in I, \quad (2)$$

where

$$\Delta_{1-\alpha}(x_j; I) := \left(\frac{\nu_0 \hat{\sigma}_j^2(x_j)}{nh_j \hat{p}_j(x_j)} \right)^{1/2} \cdot \left(\frac{C_\alpha}{A_{nj}(I)} + B_{nj}(I) \right), \quad x_j \in I.$$

Uniform Testing and Simultaneous Confidence Band

For any $x_j \in (0, 1)$, let $\hat{\sigma}_j^2(x_j) := \hat{\alpha}_j$, where

$$(\hat{\alpha}_j, \hat{\beta}_j) := \operatorname{argmin}_{\alpha, \beta \in \mathbb{R}} n^{-1} \sum_{i=1}^n K_{h_j}(x_j, X_{ij}) \{ \hat{r}_{ij} - \alpha - \beta(X_{ij} - x_j) \}^2,$$

with $\hat{r}_{ij} := \{Y_i - (\bar{Y} + \sum_{j=1}^d \hat{m}_j(X_{ij}))\}^2$ for $1 \leq i \leq n$.

Uniform Testing and Simultaneous Confidence Band

Sketch of proof of Theorem 3

Step 1: $\hat{m}_j^{\text{BR}}(x_j) - m_j(x_j)$ to weighted sum of ε

We have

$$\begin{aligned}\hat{m}_j^{\text{BR}}(x_j) - m_j(x_j) &= \hat{m}_j(x_j; \mathbf{h}) - \hat{\beta}_{j,L}(x_j)h_j^2 - m_j(x_j) \\ &= \frac{1}{nh_j p_j(x_j)} \sum_{i=1}^n K\left(\frac{x_j - X_{ij}}{h_j}\right) \varepsilon_i + R_{nj}^{\text{BR}}(x_j), \quad x_j \in I.\end{aligned}$$

Here, the remainder term R_{nj}^{BR} satisfies

$$\sup_{x_j \in I} |R_{nj}^{\text{BR}}(x_j)| = O_p\left(\sum_{k=1}^d h_k^3 + \sqrt{\frac{\log n}{n}} + \sqrt{\frac{\log n}{nh_{j,\text{BR}}}} h_j^2\right) = o_p\left(\sqrt{\frac{\log n}{nh_j}}\right),$$

Uniform Testing and Simultaneous Confidence Band

Step 2: Weighted sum of ε to weighted sum of normal random variables

Lemma 1

Assume that the conditions in hold. Then, for each $1 \leq j \leq d$, there exist random variables $Z_{1j}, \dots, Z_{nj}$ such that

$$Z_{ij} \mid X_{ij} \sim N\{0, \sigma_j^2(X_{ij})\}, \quad i = 1, \dots, n,$$

$Z_{1j}, \dots, Z_{nj}$ are independent conditionally on $X_{1j}, \dots, X_{nj}$, and for any closed interval $I \subset (0, 1)$ and $\# \in \{\text{BR}, \text{WBR}\}$, it holds that

$$\begin{aligned} \sup_{x_j \in I} \left| \frac{\sqrt{nh_j p_j(x_j)}}{\sigma_j(x_j)} \{ \hat{m}_j^\#(x_j) - m_j(x_j) \} - \frac{1}{\sqrt{nh_j p_j(x_j)} \sigma_j(x_j)} \sum_{i=1}^n K\left(\frac{x_j - X_{ij}}{h_j}\right) Z_{ij} \right| \\ = o_p(1/\sqrt{\log n}). \end{aligned}$$

Uniform Testing and Simultaneous Confidence Band

Step 3: Weighted sum of normal random variables to Gaussian process with similar covariance structure

Lemma 2

Assume that the conditions in hold. For any closed interval $I \subset (0, 1)$, let $W_{nj} \equiv W_{nj}(\cdot; I)$ be a centered stationary Gaussian process on I with covariance function

$$\text{Cov}\{W_{nj}(x_j), W_{nj}(x'_j)\} = (K * K)\left(\frac{x_j - x'_j}{h_j}\right), \quad x_j, x'_j \in I,$$

*where $(K * K)(t) := \int K(u)K(u + t) du$. Then, it holds that*

$$\sup_{t \in \mathbb{R}} \left| \mathbb{P}\left(\sup_{x_j \in I} |G_{nj}(x_j)| \leq t \mid X_{1j}, \dots, X_{nj}\right) - \mathbb{P}\left(\sup_{x_j \in I} |W_{nj}(x_j)| \leq t\right) \right| = o_p(1).$$

where

$$G_{nj}(x_j) := \frac{1}{\sqrt{nh_j p_j(x_j)} \sigma_j(x_j)} \sum_{i=1}^n K\left(\frac{x_j - X_{ij}}{h_j}\right) Z_{ij}, \quad x_j \in [0, 1],$$

where $Z_{1j}, \dots, Z_{nj}$ are the normal random variables constructed in Lemma 1.

Simulation

The covariates are generated as $X_j := \Phi(Z_j)$ for $j = 1, 2$, where $\Phi : \mathbb{R} \rightarrow [0, 1]$ denotes the cumulative distribution function of the standard normal distribution, and $(Z_1, Z_2)^\top$ is a bivariate normal random vector with $\mathbb{E}(Z_1) = \mathbb{E}(Z_2) = 0$ and $\text{Cov}(Z_j, Z_k) = 0.5^{|j-k|}$.

We used following two-dimensional additive model

$$Y = m(X_1, X_2; \rho_1, \rho_2) + \sigma(X_1, X_2) \cdot \varepsilon,$$

where

$$\begin{aligned} m(x_1, x_2; \rho_1, \rho_2) := & \sin(2\pi x_1) + \rho_1 \exp \{ -16(4x_1 - 2)^2 \} + \cos(2\pi x_2) \\ & + \rho_2 \cdot [0.3 \exp \{ -16(4x_2 - 1)^2 \} + 0.7 \exp \{ -16(4x_2 - 3)^2 \}]. \end{aligned}$$

Simulation

We first obtain the optimal bandwidth $h_{j,\text{opt}}$ for each j using the coordinate-wise bandwidth selection (CBS) algorithm.

$$h_j^{(\ell)} := \operatorname{argmin}_{g_j \in \mathcal{G}_j} \operatorname{CV}(h_1^{(\ell)}, \dots, h_{j-1}^{(\ell)}, g_j, h_{j+1}^{(\ell-1)}, \dots, h_d^{(\ell-1)}), \quad 1 \leq j \leq d, \ell \geq 1,$$

where $\mathcal{G}_j := \{h_{j,\min} + 0.005 \cdot s : s = 1, \dots, 10\}$ and $h_{j,\min}$ is the minimal bandwidth chosen to satisfy $\inf_{x_j \in [0,1]} \lambda_{\min}(\hat{\mathbf{M}}_{jj}(x_j)) > 0$.

For the auxiliary bandwidths, let $h_{\ell j} = (1 + C_\ell/10)h_{j,\text{opt}}$ for $\ell = 1, \dots, 11$, where $C_\ell = \ell - 6$.

As the initial tuples, we use $\hat{\mathbf{m}}_j^{[0],\text{tp}} = \tilde{\mathbf{m}}_j^{\text{tp}}(x_j) - \bar{Y} \cdot \mathbf{u}_0$ that satisfies the constraint $\mathbf{u}_0^\top \cdot \int_0^1 \hat{\mathbf{M}}_{jj}(x_j) \hat{\mathbf{m}}_j^{[0],\text{tp}}(x_j) dx_j = 0$.

We terminate the SBF algorithm when $\|\hat{\mathbf{m}}^{[\ell],\text{F,tp}} - \hat{\mathbf{m}}^{[\ell-1],\text{F,tp}}\|_{2,n} < \epsilon = 10^{-5}$.

Simulation

We compared the proposed estimators with two types of parametrically-helped estimators

$$g_1^{\text{oracle}}(x_1) = \sin(2\pi x_1) + \rho_1 \exp \{ -16(4x_1 - 2)^2 \},$$
$$g_2^{\text{oracle}}(x_2) = \cos(2\pi x_2) + \rho_2 [0.3 \exp \{ -16(4x_2 - 1)^2 \} + 0.7 \exp \{ -16(4x_2 - 3)^2 \}].$$

$$g_1^{\text{naive}}(x_1) = \sin(2\pi x_1), \quad g_2^{\text{naive}}(x_2) = \cos(2\pi x_2).$$

Simulation - estimation

The ISB, IV and MISE for m are defined as

$$\text{ISB} := \int_{[0,1]^2} \left(M^{-1} \sum_{i=1}^M \hat{m}^{(i)}(\mathbf{x}) - m(\mathbf{x}) \right)^2 d\mathbf{x},$$

$$\text{IV} := \int_{[0,1]^2} M^{-1} \sum_{i=1}^M \left(\hat{m}^{(i)}(\mathbf{x}) - M^{-1} \sum_{i=1}^M \hat{m}^{(i)}(\mathbf{x}) \right)^2 d\mathbf{x},$$

$$\text{MISE} := \text{ISB} + \text{IV}.$$

Similarly, for $j = 1, 2$, let $\hat{m}_j^{(i)}$ denote an estimator of m_j obtained from the i -th Monte Carlo replication. The corresponding criteria for m_j are given by

$$\text{ISB}_j := \int_0^1 \left(M^{-1} \sum_{i=1}^M \hat{m}_j^{(i)}(x_j) - m_j(x_j) \right)^2 dx_j,$$

$$\text{IV}_j := \int_0^1 M^{-1} \sum_{i=1}^M \left(\hat{m}_j^{(i)}(x_j) - M^{-1} \sum_{i=1}^M \hat{m}_j^{(i)}(x_j) \right)^2 dx_j,$$

Simulation - estimation

Table 1: The values of ISB, IV and MISE, multiplied by 10^3 , for the local linear SBF estimators, two PH estimators, and the proposed biased reduced estimators.

(ρ_1, ρ_2)	n		$\hat{m}^{LL}$	$\hat{m}_{\text{oracle}}^{PH}$	$\hat{m}_{\text{naive}}^{PH}$	$\bar{m}^{BR}$	$\bar{m}^{WBR}$	$\hat{m}^{BR}$	$\hat{m}^{WBR}$
(0,3)	400	ISB	2.99	2.07	2.33	2.18	2.15	2.14	2.13
		IV	2.78	4.25	4.35	3.96	3.76	3.63	3.53
		MISE	5.77	6.32	6.68	6.15	5.92	5.77	5.65
	700	ISB	1.16	0.55	0.59	0.50	0.48	0.46	0.45
		IV	1.99	2.93	2.97	2.57	2.49	2.43	2.39
		MISE	3.15	3.47	3.55	3.07	2.97	2.90	2.84
	1000	ISB	1.07	0.36	0.43	0.35	0.34	0.34	0.33
		IV	1.50	1.85	1.95	1.68	1.63	1.60	1.57
		MISE	2.57	2.21	2.38	2.04	1.97	1.93	1.90
	400	ISB	2.13	1.09	1.26	1.02	0.98	0.96	0.94
		IV	2.98	4.94	5.00	4.34	4.09	3.92	3.79
		MISE	5.11	6.03	6.25	5.36	5.06	4.88	4.73
(2,0)	700	ISB	2.08	0.66	0.79	0.66	0.65	0.64	0.64
		IV	2.22	3.03	3.11	2.73	2.64	2.58	2.53
		MISE	4.30	3.69	3.91	3.39	3.29	3.22	3.17
	1000	ISB	1.73	0.40	0.49	0.38	0.38	0.39	0.39
		IV	1.71	2.11	2.09	1.85	1.80	1.76	1.74
		MISE	3.44	2.51	2.58	2.24	2.18	2.15	2.13

(a) whole function

Table 2: The values of ISB₁, IV₁ and MISE₁, multiplied by 10^3 , for the local linear SBF estimators, two PH estimators, and the proposed biased reduced estimators.

(ρ_1, ρ_2)	n		$\hat{m}_1^{LL}$	$\hat{m}_{1,\text{oracle}}^{PH}$	$\hat{m}_{1,\text{naive}}^{PH}$	$\bar{m}_1^{BR}$	$\bar{m}_1^{WBR}$	$\hat{m}_1^{BR}$	$\hat{m}_1^{WBR}$
(0,3)	400	ISB ₁	0.51	0.33	0.33	0.48	0.46	0.45	0.43
		IV ₁	1.93	1.87	1.87	2.67	2.56	2.49	2.44
		MISE ₁	2.44	2.20	2.20	3.15	3.02	2.94	2.87
	700	ISB ₁	0.41	0.37	0.37	0.52	0.51	0.50	0.49
		IV ₁	0.51	0.50	0.50	0.85	0.81	0.78	0.75
		MISE ₁	0.91	0.87	0.87	1.37	1.32	1.28	1.25
	1000	ISB ₁	0.14	0.06	0.06	0.13	0.13	0.12	0.11
		IV ₁	0.30	0.27	0.27	0.48	0.45	0.43	0.42
		MISE ₁	0.44	0.33	0.33	0.61	0.57	0.55	0.53
	400	ISB ₁	1.90	0.41	1.70	0.66	0.62	0.61	0.59
		IV ₁	2.17	1.69	1.76	2.90	2.75	2.66	2.58
		MISE ₁	4.07	2.10	3.46	3.56	3.37	3.27	3.18
(2,0)	700	ISB ₁	1.96	0.41	1.40	0.63	0.63	0.64	0.64
		IV ₁	1.02	0.63	0.72	1.25	1.19	1.15	1.12
		MISE ₁	2.98	1.04	2.12	1.89	1.83	1.79	1.76
	1000	ISB ₁	1.52	0.11	0.80	0.22	0.22	0.23	0.23
		IV ₁	0.73	0.47	0.46	0.72	0.68	0.66	0.66
		MISE ₁	2.25	0.57	1.26	0.94	0.91	0.89	0.89

(b) componentwise

Simulation - confidence band

We compare the proposed testing procedure with an oracle approach. Specifically, we define the oracle responses as

$$\begin{aligned} Y^{\text{oracle},1} &:= Y - m_1(X_1) = m_2(X_2) + \varepsilon, \\ Y^{\text{oracle},2} &:= Y - m_2(X_2) = m_1(X_1) + \varepsilon. \end{aligned}$$

Note that $Y^{\text{oracle},j}$ depends only on the j -th covariate X_j . Using these oracle responses, we construct an SCB for m_j over I by applying the method of [Cheng et al., 2018], denoted as $\text{SCB}_{\text{oracle}}$.

Simulation - confidence band

Coverage rates(%) of the SCBs under different variance levels and sample sizes.

n	α	m_1		m_2	
		SCB _{oracle}	SCB	SCB _{oracle}	SCB
1000	0.4	94	95	95	94
	0.7	93	97	92	98
	1.0	96	97	91	97
1500	0.4	96	94	97	93
	0.7	98	96	95	97
	1.0	94	97	93	97

Bike rental data

To assess predictive performance, we adopt 5-fold cross-validation and compute the average squared prediction error (ASPE) defined by

$$\text{ASPE} = \frac{1}{n} \sum_{k=1}^5 \sum_{i \in I_k} (Y_i - \hat{Y}_i^{(-I_k)})^2,$$

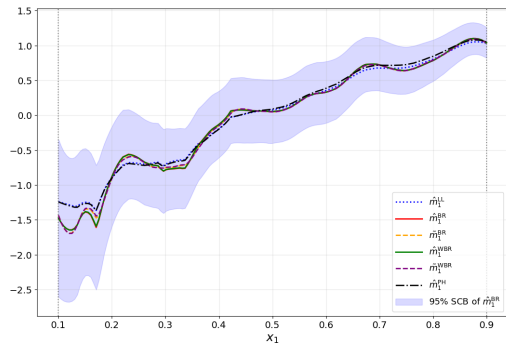
where I_k denotes the index set of the k -th fold and $\hat{Y}_i^{(-I_k)}$ denotes the prediction of Y_i obtained from the model trained without the observations in I_k . Table reports the 5-fold cross-validation ASPE values for the competing estimators.

Estimator	$\hat{m}^{\text{LL}}$	$\hat{m}^{\text{PH}}$	$\tilde{m}^{\text{BR}}$	$\tilde{m}^{\text{WBR}}$	$\hat{m}^{\text{BR}}$	$\hat{m}^{\text{WBR}}$
ASPE	1.7630	1.7628	1.7459	1.7457	1.7453	1.7451

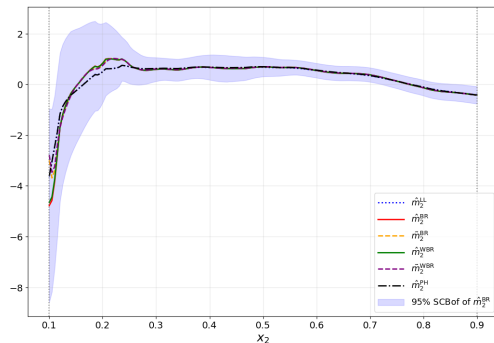
$n = 941$ observations.

Bike rental data

The auxiliary bandwidths are chosen as $h_{\ell j} = (1 + C_\ell/20)h_{j,\text{opt}}$, where $C_\ell = 1, 2, \dots, 7$.



(a) The effect of temperature



(b) The effect of humidity

The test statistics $T_{nj}(I)$ are 27.67 and 17.02 for the temperature and humidity, respectively. P-values are 1.93×10^{-12} and 8.11×10^{-8} , respectively. These results indicate that both temperature and humidity have statistically significant effects on bike rental counts at the 5% significance level.

Flights data

$n = 978$ observations.

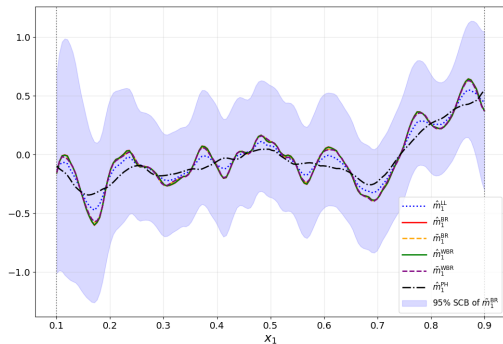
The auxiliary bandwidths are chosen as $h_{\ell j} = (1 + C_\ell/20)h_{j,\text{opt}}$, where $C_\ell = 1, 2, \dots, 7$.

Estimator	$\hat{m}^{\text{LL}}$	$\hat{m}^{\text{PH}}$	$\tilde{m}^{\text{BR}}$	$\tilde{m}^{\text{WBR}}$	$\hat{m}^{\text{BR}}$	$\hat{m}^{\text{WBR}}$
ASPE	1.5500	1.5488	1.5483	1.5488	1.5487	1.5478

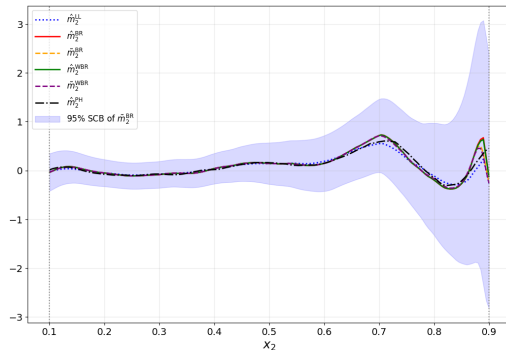
Table: Comparison of 5-fold cross-validation ASPE values in the analysis of the flights dataset.

For the PH estimator, we employ $g_1(x_1) = \sin(2\pi x_1)$ and $g_2(x_2) = \cos(2\pi x_2)$ as the prespecified functions.

Flights data



(a) The effect of temperature



(b) The effect of wind speed

The test statistics $T_{nj}(I)$ are 5.74 and 3.48 for temperature and wind speed, respectively. P-values are 0.0064 and 0.0597 respectively. The null hypothesis is rejected for temperature but is not rejected for wind speed. These results provide evidence that temperature has a significant effect on flight delays at the 5% significance level, whereas the effect of wind speed is not statistically significant.



Cheng, M.-Y., Huang, T., Liu, P., and Peng, H. (2018).

Bias reduction for nonparametric and semiparametric regression models.

Statistica Sinica, 28(4):2749–2770.



Hong, H., Lee, Y. K., and Park, B. U. (2023).

Additive regression with parametric help.

Bernoulli, 29(4):3059–3092.



Jeon, J. M., Lee, Y. K., Mammen, E., and Park, B. U. (2022).

Locally polynomial hilbertian additive regression.

Bernoulli, 28(3):2034–2066.



Lee, Y. K., Mammen, E., Nielsen, J. P., and Park, B. U. (2020).

Nonparametric regression with parametric help.

Electron. J. Stat., 14(2):3845–3868.